

**University of North Carolina at Charlotte
Department of Electrical and Computer Engineering**

Homework Report 4

ECGR 5106 - Introduction to Deep Learning

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GitHub: <https://github.com/vipravlipare/ECGR-5106-Intro-To-Deep-Learning>

Homework 4 Report

Overall Setup and Assumptions

All experiments used PyTorch and CPU hardware. Problems 1 and 2 were character-level next-character prediction tasks. Problems 3 and 4 used the same `vast_english_french.txt` split from Homework 3, with 444 training sentence pairs and 111 validation sentence pairs. The main validation metrics were calculated only on held-out validation data.

For translation, Traditional Sequence Accuracy means exact sequence accuracy, where the whole predicted sentence must match the target exactly. BLEU-4 is also reported because it gives partial credit when the generated translation has overlapping word or character patterns with the reference. Every time column in this report is total training time or inference time, and the complexity column is an approximate FLOPs/operation-count proxy from the notebook.

The submitted GitHub folder should include the executed Jupyter notebooks with visible cell outputs. At the time this report was checked, the four Homework 4 notebooks had saved outputs and no saved error cells.

Problem 1: Transformer Character Prediction on Long Sequence

Objective

For this problem, a Transformer model was trained on the same long character sequence used in Homework 2. The goal was to compare sequence lengths 10, 20, and 30, then compare the Transformer results against the Homework 2 recurrent models.

Model Setup

The Problem 1 model used a character-level Transformer encoder for next-character prediction. Each character was converted into an embedding, positional encoding was added to preserve character order, and the output went through a final linear layer to predict the next character. The final model used `hidden_size = 32`, 1 Transformer block, and 2 attention heads. This was kept small because training was done on CPU, but it still allowed the model to learn strong character patterns. The main advantage over the Homework 2 RNN, LSTM, and GRU models is that self-attention can compare multiple character positions at the same time instead of processing the sequence one step at a time.

Training Setup and Hyperparameters

The model was trained on sequence lengths 10, 20, and 30, as required by the rubric. The runs used AdamW, learning rate 0.0025, weight decay 0.001, dropout 0.20, batch size 128, validation batch size 256, and CrossEntropyLoss with light label smoothing. ReduceLROnPlateau lowered the learning rate when validation performance stopped improving, and early stopping helped prevent unnecessary extra training. The maximum epoch count was 250, but each sequence length stopped at a different point depending on validation performance.

Problem 1 Results

Seq	Epochs	Train Loss	Val Loss	Best Val Acc	Top-5 Acc	Total Train Time
10	138	0.6620	0.5879	91.27%	96.40%	6.51 sec

20	125	0.5960	0.4731	94.67%	97.70%	9.32 sec
30	152	0.5864	0.4028	96.60%	98.58%	17.23 sec

Seq	Last-Step Acc	Params	Size MB	Approx FLOPs Proxy
10	26.67%	11.5k	0.044	26.9k
20	24.67%	11.5k	0.044	66.6k
30	17.33%	11.5k	0.044	119.0k

The best Problem 1 Transformer result came from sequence length 30. This model reached 96.60% best validation accuracy with about 17.23 seconds of total CPU training time. This was the strongest result because the longer sequence gave the Transformer more context before predicting the next character. The sequence length 10 and 20 models also learned the task well, but the sequence length 30 model had the best overall validation performance.

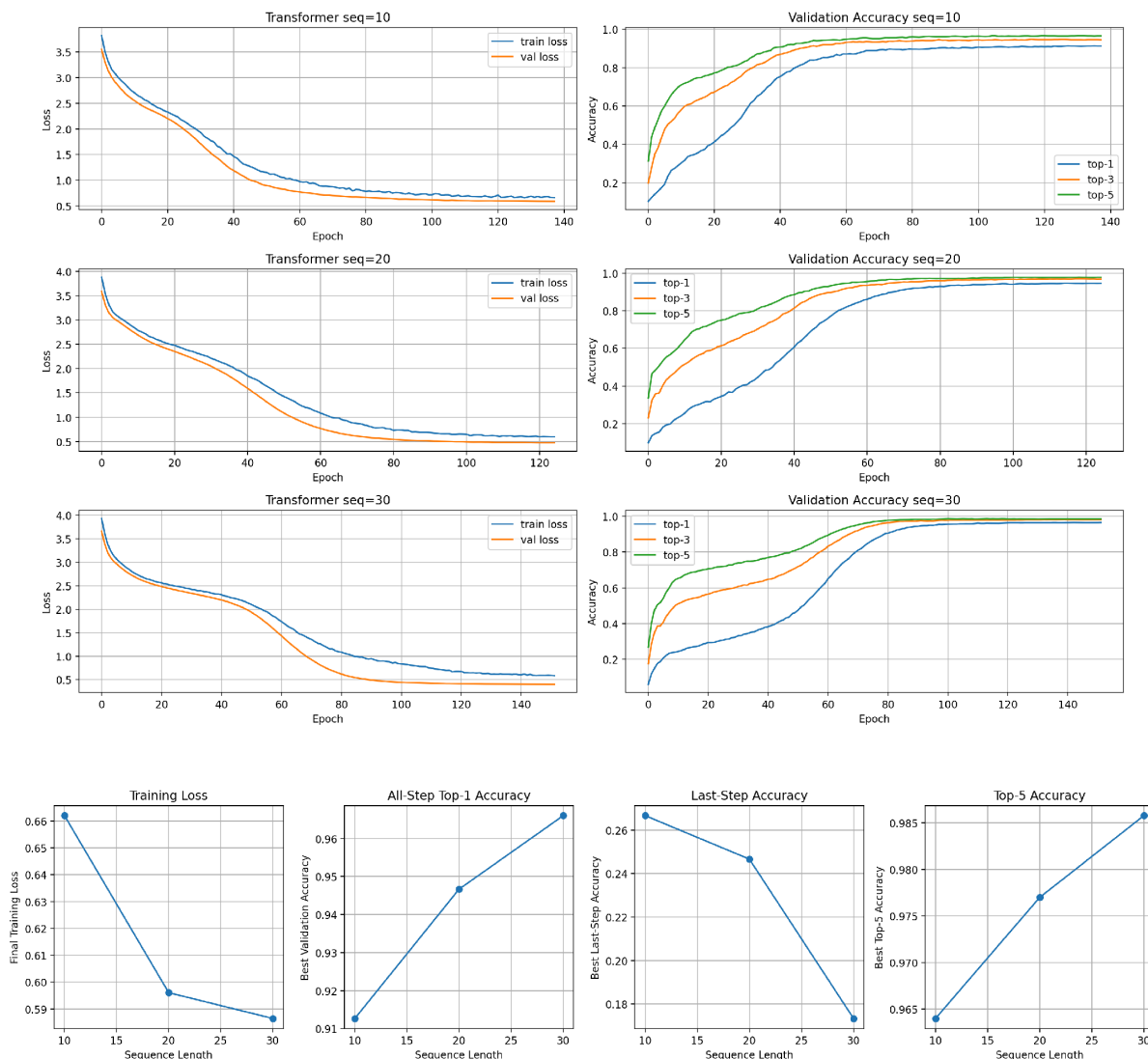
For the sequence-length result table, the main trend was that validation accuracy improved as the sequence length increased. This makes sense because the model had more previous characters available when making each prediction. The sequence length 30 model also had a slightly higher computational cost than the shorter sequence models, but the increase was still small because the model only used one Transformer block and a hidden size of 32.

Source	Model	Seq	Train / Val Loss	Best Val Acc	Top-5 Acc	Total Time	Params / FLOPs
HW2	GRU	10	1.0417 / 1.0654	77.21%	94.84%	77.20s	110.6k / 983.0k
HW2	LSTM	10	1.0284 / 1.0773	76.92%	94.74%	75.19s	143.7k / 1.31M
HW2	RNN	10	1.1546 / 1.1691	74.19%	94.42%	53.59s	44.6k / 327.7k
HW2	GRU	20	0.7811 / 0.7568	88.31%	97.49%	185.64s	110.6k / 1.97M
HW2	LSTM	20	0.8215 / 0.7880	87.79%	97.26%	143.06s	143.7k / 2.62M
HW2	RNN	20	0.8872 / 0.8484	86.53%	97.02%	95.72s	44.6k / 655.4k
HW2	GRU	30	0.7206 / 0.6654	92.10%	98.23%	204.06s	110.6k / 2.95M
HW2	LSTM	30	0.7381 / 0.6851	91.72%	98.13%	238.00s	143.7k / 3.93M
HW2	RNN	30	0.7877 / 0.7366	90.42%	97.95%	172.26s	44.6k / 983.0k
HW4	Transformer	10	0.6620 / 0.5879	91.27%	96.40%	6.51s	11.5k / 26.9k
HW4	Transformer	20	0.5960 / 0.4731	94.67%	97.70%	9.32s	11.5k / 66.6k
HW4	Transformer	30	0.5864 / 0.4028	96.60%	98.58%	17.23s	11.5k / 119.0k

This comparison shows the biggest difference between Homework 2 and Homework 4 for the long sequence task. In Homework 2, increasing the sequence length helped every recurrent model, and the GRU sequence length 30 model was the best RNN-based result with 92.10% best validation accuracy. In Homework 4, the Transformer sequence length 30 model improved this to 96.60% while using only 11.5k parameters and a 119.0k FLOPs proxy. The recurrent models had to process characters one step at a time, so their training time and complexity

increased more as the sequence length grew. The Transformer used self-attention to compare positions in the sequence directly, which made it much faster and more accurate for this smaller long-sequence prediction task. The tradeoff is that Homework 2 recurrent models had much stronger last-step behavior, while the Homework 4 Transformer was strongest on all-step validation accuracy.

For the Homework 2 comparison table, the Transformer is compared against all Homework 2 Problem 1 recurrent runs: RNN, LSTM, and GRU at sequence lengths 10, 20, and 30. This is stronger than only comparing against one best recurrent row. The main trend is that the Homework 4 Transformer had the highest best validation accuracy at each matching sequence length, especially at sequence length 30 where it reached 96.60%. The best Homework 2 recurrent model was GRU sequence length 30 with 92.10% best validation accuracy. The Transformer also used fewer parameters and a much smaller FLOPs proxy than the LSTM and GRU models. Cross-attention was not part of the Homework 2 next-character prediction models, so this comparison is against the RNN, LSTM, and GRU recurrent baselines using the metrics available from that homework.



The training and validation loss plots show that the model learned steadily over time. Training loss decreased as expected, and validation loss also improved instead of rising sharply. This means the model did not show major harmful overfitting for this short-text task. The validation accuracy plot also supports this because accuracy increased during training and stayed strong by the end. The summary comparison plot shows that the sequence length 30 model was the best overall, while sequence lengths 10 and 20 were still useful but slightly weaker.

The main weakness was last-step accuracy. Even though the all-step validation accuracy was high, the final-character prediction and free-generation behavior were weaker. This means the model learned the training/validation prediction task well, but it could still become repetitive when generating characters without the true previous targets. Overall, the Transformer was the best model for Problem 1 because it reached high validation accuracy, trained quickly on CPU, and stayed smaller than the Homework 2 recurrent models.

Problem 2: Transformer for Tiny Shakespeare

Objective

For this problem, a Transformer was trained on Tiny Shakespeare for next-character prediction. The starting model was a 2-block, 2-head Transformer. The model was trained for sequence lengths 20 and 30, then a longer sequence length of 50 was tested.

Model Setup

The Problem 2 model used a character-level Transformer for Tiny Shakespeare. Each character was passed through an embedding layer, positional encoding was added, and stacked self-attention blocks were used before the final classifier. The starting configuration was the required 2-block, 2-head Transformer. The Transformer table reports the block/head study for blocks = 1, 2, 4 and heads = 2, 4, plus the required sequence length 30 baseline run and the sequence length 50 run. The Homework 2 comparison table includes the earlier LSTM and GRU base runs, smaller hidden-size runs, two-layer runs, and sequence length 50 runs, so the comparison is against the full RNN-based set instead of only one LSTM row.

Training Setup and Hyperparameters

The runs used 25 epochs, AdamW, learning rate = 0.0035, dropout = 0.08, batch size = 128, CrossEntropyLoss, and ReduceLROnPlateau. Training used a maximum of 55 training batches per epoch and 20 validation batches because the full Tiny Shakespeare run was very slow on CPU. The report includes training loss, validation loss, top-1 accuracy, top-5 accuracy, total training time, inference time, generated text, model size, and the approximate FLOPs proxy. Perplexity is $\exp(\text{validation loss})$, so lower perplexity means the model is less uncertain. Hidden size was also treated as part of the hyperparameter study. The main starting configuration used 2 Transformer blocks and 2 attention heads, but the later experiments changed the number of blocks, number of heads, and hidden size to see how model capacity affected validation accuracy, validation loss, runtime, perplexity, and model complexity. This was important because the Transformer can become slow or overfit if the hidden size and number of blocks are increased too much for the Tiny Shakespeare character-prediction task.

Problem 2 Results

Run	Seq	Blocks	Heads	Hidden	Train Loss	Val Loss	Top-1 Acc	PPL	Total Train Time
arch_s20	20	1	2	96	2.0284	1.9925	41.95%	7.33	75.99 sec
arch_s20	20	1	4	128	1.9342	1.9367	43.55%	6.94	466.30 sec
arch_s20	20	2	2	96	1.9164	1.9248	44.69%	6.85	228.30 sec
arch_s20	20	2	4	128	1.9436	1.9498	42.62%	7.03	252.22 sec
arch_s20	20	4	2	96	1.9640	1.9543	43.20%	7.06	316.34 sec
arch_s20	20	4	4	128	2.0770	2.0367	41.13%	7.67	515.91 sec
base_s30	30	2	2	96	1.9758	1.8459	45.51%	6.33	289.96 sec
seq50	50	2	4	128	2.0163	2.0228	40.31%	7.56	1264.92 sec

For the main Problem 2 result table, the best saved Transformer run was base_s30. This model used sequence length 30, 2 blocks, 2 heads, and hidden size 96. It reached 45.51% top-1 validation accuracy, 78.87% top-5 validation accuracy, 1.8459 validation loss, and 6.33 perplexity. This was the best Transformer result because it had the strongest validation accuracy and the lowest validation loss/perplexity. For the sequence length 20 runs, the 2-block, 2-head model was the strongest. This shows that moderate depth helped, but increasing to 4 blocks did not improve validation accuracy enough to justify the extra time and parameters.

For sequence length 50, the model had more context available, but the extra context did not automatically improve performance. The longer sequence increased training time and computational cost because self-attention compares positions across the sequence. The perplexity and validation accuracy showed that the model learned useful character patterns, but the sequence length 50 run was not the best overall tradeoff. This suggests that, for this CPU-based experiment, a moderate sequence length was more practical than simply making the context window longer.

Run	Seq	Params	Size MB	Approx FLOPs	Inference	Generated Output Snippet
arch_s20	20	86.6k	0.33	445.4k	0.16 sec	To be, or not to be well the whave shin his the the to spars andst sou w...
arch_s20	20	148.2k	0.57	757.8k	0.19 sec	To be, or not to be than ta mase the sin to the to hear the more henow s...
arch_s20	20	161.3k	0.62	890.9k	0.23 sec	To be, or not to be come tan the wath her that them. CORIOLANUS: Wher h...
arch_s20	20	280.6k	1.07	1.52M	0.26 sec	To be, or not to bere to the sine to here the with the the his the tid a...
arch_s20	20	310.9k	1.19	1.78M	0.40 sec	To be, or not to be had and to shall the are an though their as to hang ...
arch_s20	20	545.6k	2.08	3.03M	0.68 sec	To be, or not to be the found thate ole at and the at tor to his there h...
base_s30	30	161.3k	0.62	1.45M	0.31 sec	To be, or not to be and they thee the tand tell the arits The ane mested...
seq50	50	280.6k	1.07	4.56M	0.49 sec	To be, or not to bearet hound how the warde to thound a the with wathill...

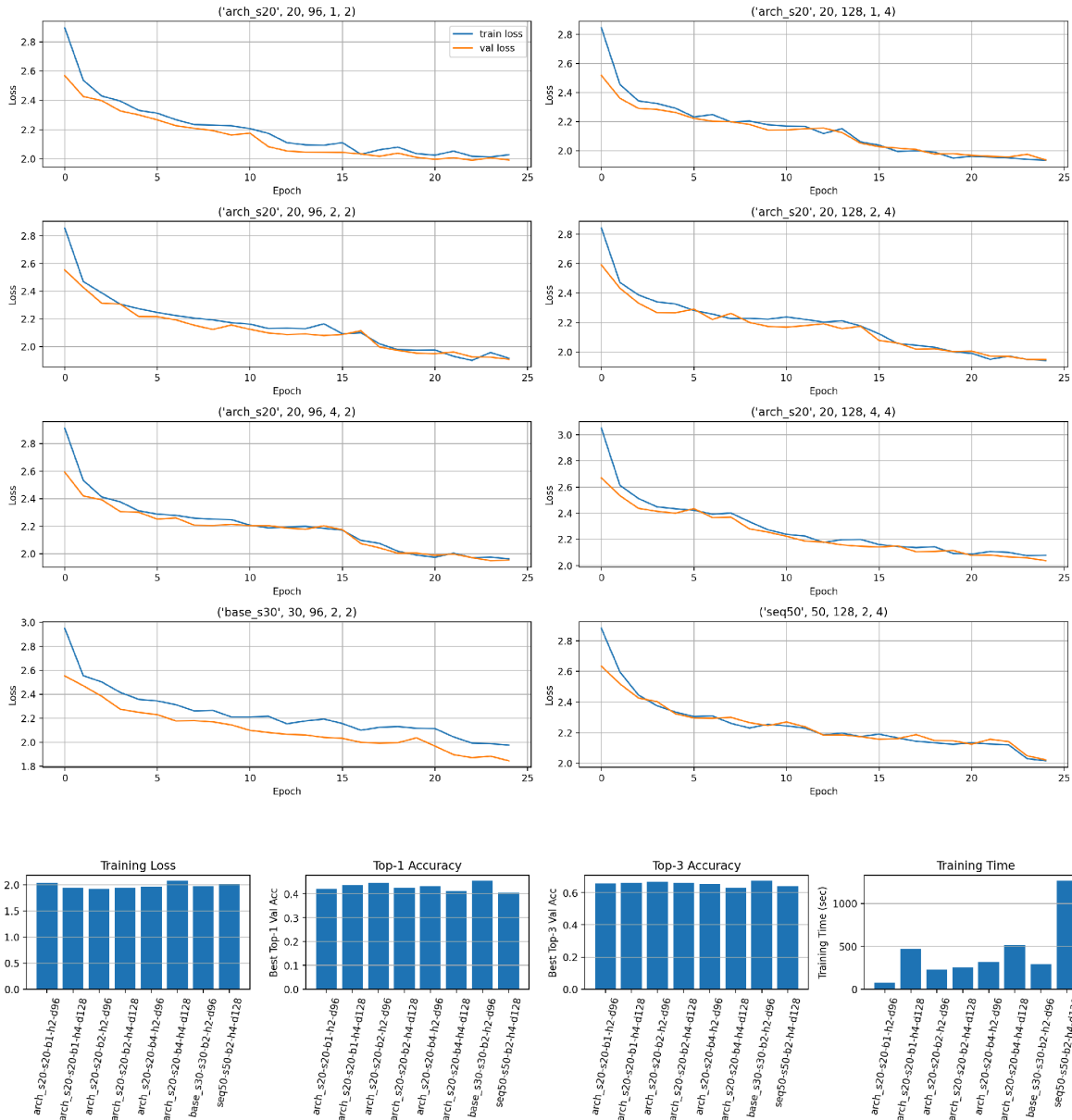
For the complexity and output table, the larger Transformer models had more parameters, higher FLOPs, and longer inference time. The expanded Homework 2 comparison shows that

the strongest recurrent result was the LSTM sequence length 20 run when validation loss, top-5 accuracy, and perplexity are considered together. That Homework 2 LSTM had 52.77% best validation accuracy, 83.39% top-5 accuracy, and 4.78 perplexity. The best Homework 4 Transformer reached 45.51% top-1 accuracy, 78.87% top-5 accuracy, and 6.33 perplexity. The Homework 2 GRU sequence length 20 had the slightly highest final top-1 accuracy at 52.62%, so both LSTM and GRU beat the Transformer on Tiny Shakespeare top-1 accuracy. The sequence length 50 Transformer was much slower, taking 1264.92 seconds, but it only reached 40.31% top-1 accuracy and 7.56 perplexity. This means the longer Transformer context increased computation more than it improved learning.

The expanded Homework 2 rows are important because Homework 2 did not only test one LSTM. It tested LSTM and GRU models at sequence lengths 20 and 30, smaller hidden size models, two-layer models, and sequence length 50 models. The Homework 4 Transformer is therefore compared against the full earlier RNN-based set, not just against the single best row.

Source/Run	Model	Seq	Top-1	Top-5	Val Loss	PPL	Train / Infer Time	Params	FLOPs
HW2 base	LSTM	20	52.46%	83.39%	1.5646	4.78	193.37s / 0.13s	164.5k	2.62M
HW2 base	GRU	20	52.62%	82.64%	1.5879	4.89	471.89s / 0.26s	131.5k	1.97M
HW2 base	LSTM	30	52.53%	83.27%	1.5802	4.86	238.55s / 0.14s	164.5k	3.93M
HW2 base	GRU	30	52.02%	82.37%	1.6120	5.01	280.12s / 0.31s	131.5k	2.95M
HW2 small	LSTM	20	50.98%	82.04%	1.6539	5.23	153.92s / 0.08s	45.4k	655.4k
HW2 small	GRU	20	50.97%	81.59%	1.6571	5.24	247.97s / 0.22s	37.1k	491.5k
HW2 2-layer	LSTM	20	52.45%	82.88%	1.5755	4.83	443.62s / 0.17s	296.6k	5.24M
HW2 2-layer	GRU	20	51.56%	82.37%	1.6175	5.04	537.52s / 0.44s	230.6k	3.93M
HW2 seq50	LSTM	50	52.30%	83.42%	1.5680	4.80	293.13s / 0.18s	164.5k	6.55M
HW2 seq50	GRU	50	52.02%	82.81%	1.5918	4.91	318.01s / 0.53s	131.5k	4.92M
HW4 best	Transformer	30	45.51%	78.87%	1.8459	6.33	289.96s / 0.31s	161.3k	1.45M
HW4 seq50	Transformer	50	40.31%	N/A	2.0228	7.56	1264.92s / 0.49s	280.6k	4.56M

This expanded Tiny Shakespeare comparison gives a more complete picture than only comparing against one LSTM row. Homework 2 tested base LSTM and GRU models, smaller hidden-size models, two-layer models, and sequence length 50 models. Across those runs, the LSTM sequence length 20 model was the strongest overall language model because it had 52.77% best validation accuracy, 83.39% top-5 accuracy, and 4.78 perplexity. The Homework 2 GRU sequence length 20 model had the slightly highest final top-1 accuracy at 52.62%, so both recurrent families stayed stronger than the Homework 4 Transformer on Tiny Shakespeare. The best Homework 4 Transformer reached 45.51% top-1 accuracy and 6.33 perplexity, which means it learned useful Shakespeare-style patterns but did not match the recurrent models on strict next-character prediction. The Transformer still had a lower FLOPs proxy than many of the larger Homework 2 recurrent runs, but its longer sequence length 50 run became slow and did not improve accuracy, so more context was not automatically better.



The Problem 2 plots support the same conclusion as the tables. Training loss decreased during the runs, so the model was learning the Tiny Shakespeare text. However, validation accuracy and perplexity did not always improve when the model became larger. The loss curves show that the stronger Transformer runs reduced validation loss better, while the summary plot shows the tradeoff between accuracy, perplexity, time, parameters, and FLOPs. Overall, the plots show that the moderate 2-block setup was the best Transformer balance, while the largest and longest-context runs were slower without giving better validation results. Compared with the full Homework 2 table, the Transformer was competitive in model size but weaker than the best LSTM/GRU runs on top-1 accuracy and perplexity. This also satisfies the rubric requirement to discuss running time and computational perplexity because the model choice is explained using accuracy, validation loss, perplexity, total time, model size, and FLOPs instead of accuracy alone.

Problem 3: Transformer-Based English-to-French Translation

Objective

For this problem, the Homework 3 English-to-French translation task was repeated with a Transformer encoder-decoder model. The same 80/20 train-validation split from Homework 3 was used so that the Transformer could be compared fairly against the GRU baseline and GRU attention model.

Model Setup

The Problem 3 model used a Transformer encoder-decoder architecture for English-to-French translation. The source English sentence and target French sentence were both treated as character sequences. The model used source and target embeddings, positional encoding, Transformer encoder-decoder layers, and a final linear layer to predict the next target character. The required 1-block, 2-block, and 4-block settings with 2 and 4 heads are included, and a 3-block middle case is also shown as an extra depth comparison.

Training Setup and Hyperparameters

The runs used hidden size 128, dropout 0.05, Adam, learning rate = 0.004, weight decay = 0.00001, batch size = 96, CrossEntropyLoss with padding ignored, ReduceLROnPlateau, and up to 35 epochs on CPU. Padding was ignored so the model was not rewarded or punished for predicting PAD tokens. The table reports training loss, validation loss, exact sequence accuracy, word-level BLEU-4, character-level BLEU-4, total training time, parameters, and the approximate FLOPs proxy. The assignment metric is word-level BLEU-4, while character BLEU is included because the model itself is character-level.

Problem 3 Results

Config	Blocks	Heads	Train Loss	Val Loss	Exact Acc	Word BLEU-4	Char BLEU-4
blocks1_heads2	1	2	0.0352	4.3113	0.00%	0.1300	0.3659
blocks1_heads4	1	4	0.0178	4.3493	0.00%	0.1223	0.3600
blocks2_heads2	2	2	0.2695	4.9510	0.00%	0.0701	0.2386
blocks2_heads4	2	4	0.3677	4.9814	0.00%	0.0508	0.2122
blocks3_heads2	3	2	5.1415	6.0152	0.00%	0.0079	0.0222
blocks3_heads4	3	4	5.1346	5.8443	0.00%	0.0131	0.0240
blocks4_heads2	4	2	5.4764	6.1847	0.00%	0.0000	0.0000
blocks4_heads4	4	4	5.4837	6.1982	0.00%	0.0000	0.0000

Config	Best Val Loss	Total Train Time	Params	Approx FLOPs Proxy
blocks1_heads2	3.9253	61.90 sec	1.25M	1.26M
blocks1_heads4	3.8254	37.58 sec	1.25M	1.26M
blocks2_heads2	3.9651	60.71 sec	1.71M	2.53M
blocks2_heads4	4.1028	57.35 sec	1.71M	2.53M
blocks3_heads2	5.8819	143.42 sec	2.18M	3.79M

blocks3_heads4	5.7680	53.91 sec	2.18M	3.79M
blocks4_heads2	5.8874	23.44 sec	2.64M	5.05M
blocks4_heads4	5.8770	75.61 sec	2.64M	5.05M

For the Problem 3 configuration table, the best English-to-French Transformer was blocks1_heads2. It reached 0.1300 word BLEU-4 and 0.3659 character BLEU-4. Exact sequence accuracy stayed at 0.00%, but BLEU-4 improved because several predictions shared useful words or character patterns with the target translation. The 1-block models performed better than the deeper 3-block and 4-block models. This means the best model was not the deepest model. On this small dataset, the shallow Transformer generalized better.

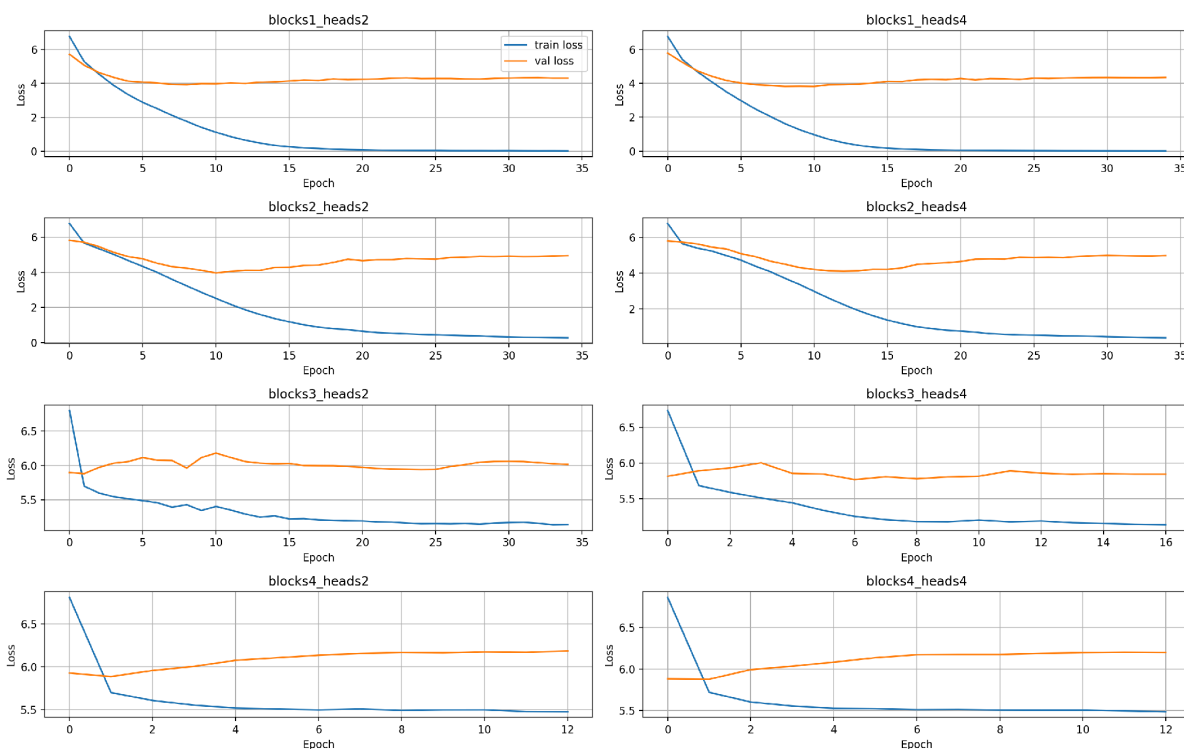
For the complexity and Homework 3 comparison tables, the best Transformer used more parameters than the Homework 3 GRU models, but it gave better BLEU-4 results. The Homework 3 baseline GRU had 0.0501 word BLEU-4, and the Homework 3 attention GRU had 0.0899 word BLEU-4. The best Homework 4 Transformer improved this to 0.1300 word BLEU-4. This means the Transformer improved the main translation-quality metric even though traditional exact sequence accuracy stayed at 0.00%. The tradeoff is that the Transformer had more parameters than the GRU models, but it was still reasonable to train on CPU for this dataset. The qualitative table supports the numeric result because the generated French translations were not exact matches, but several kept the main phrase structure. This directly compares the Transformer against the Homework 3 model without attention and the Homework 3 model with attention.

Model	Direction	Exact	Word BLEU	Char BLEU	Best Val Loss	Total Train Time	Params
Homework 3 Baseline GRU	English to French	0.00%	0.0501	0.1881	1.4115	61.77 sec	332.2k
Homework 3 Attention GRU	English to French	0.90%	0.0899	0.2620	1.4355	259.55 sec	666.9k
Homework 4 Best Transformer	English to French	0.00%	0.1300	0.3659	3.9253	61.90 sec	1.25M

Qualitative Validation

The qualitative table for Problem 3 gives several English validation inputs, the target French sentence, the predicted French sentence, exact-match status, word BLEU-4, and character BLEU-4. These examples are important because exact-match accuracy alone makes the model look like it learned nothing, even when the generated sentence is partly correct. Most predictions were not perfect full-sentence translations, but several of them kept useful parts of the target meaning. For example, some outputs kept the correct subject or phrase structure, but changed the object, ending, or extra words. This is why BLEU-4 is useful here: it gives partial credit when the predicted French sentence overlaps with the target, while exact sequence accuracy only gives credit for a perfect match.

English Input	Target French	Predicted French	Exact	Word BLEU-4	Char BLEU-4
they feed the pigeons in the sq...	ils nourrissent les pigeons sur...	ils nourrissent les oiseaux dan...	FAIL	0.1757	0.6012
she practices yoga every morning	elle pratique le yoga tous les ...	elle joue du violon tous les jours	FAIL	0.0773	0.3999
i enjoy walking in the snow	j'aime marcher dans la neige	j'aime marcher dans la forêt dense	FAIL	0.5081	0.6816
she loves to wear modern jackets	elle adore porter des vestes mo...	elle adore porter des vestes en...	FAIL	0.5411	0.7979
the city bus arrives precisely ...	le bus de la ville arrive préci...	le bus arrive à cinq heures	FAIL	0.1365	0.3847



The Problem 3 plots show a clear generalization gap. Training loss decreased strongly, but validation loss stayed much higher, especially for deeper models. This shows overfitting: the model learned the training sentence pairs better than the validation sentence pairs. The summary plot also shows that deeper models increased parameters and FLOPs, but did not improve BLEU-4. Overall, the plots match the tables because the shallow 1-block Transformer gave the best validation translation quality.

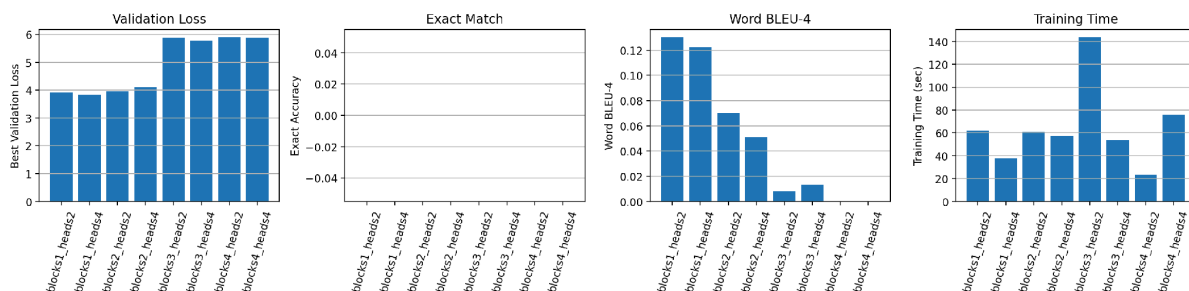


Figure: Problem 3 validation BLEU-4, exact accuracy, time, parameters, and complexity summary.

Problem 4: Reversing the Direction to French-to-English Translation

Objective

For this problem, the translation direction was reversed. French was used as the source language and English was used as the target language. The same split membership from Homework 3 was reused so that the direction comparison stayed fair.

Model Setup and Training

The Problem 4 model used the same Transformer encoder-decoder structure as Problem 3, but the language direction was reversed. French was used as the source sequence and English was used as the target sequence. Keeping the same architecture made the comparison fair because the main change was the translation direction. The required 1-block, 2-block, and 4-block models with 2 and 4 heads are included, and 3-block results are shown as an extra middle-depth comparison. The same hidden size, optimizer, dropout, batch size, loss function, scheduler, and CPU setup from Problem 3 were used.

Problem 4 Results

Config	Blocks	Heads	Train Loss	Val Loss	Exact Acc	Word BLEU-4	Char BLEU-4
blocks1_heads2	1	2	0.0386	4.7659	0.00%	0.1246	0.3411
blocks1_heads4	1	4	0.0218	4.7395	0.00%	0.1471	0.3619
blocks2_heads2	2	2	0.5405	5.5574	0.00%	0.0505	0.1639
blocks2_heads4	2	4	0.3364	5.2560	0.00%	0.0729	0.2435
blocks3_heads2	3	2	5.5483	6.3065	0.00%	0.0010	0.0003
blocks3_heads4	3	4	4.8930	5.8337	0.00%	0.0010	0.0003
blocks4_heads2	4	2	5.3663	6.1957	0.00%	0.0000	0.0000
blocks4_heads4	4	4	5.3653	6.1873	0.00%	0.0000	0.0000

Config	Best Val Loss	Total Train Time	Params	Approx FLOPs Proxy
blocks1_heads2	4.2592	51.86 sec	1.25M	1.26M
blocks1_heads4	4.1966	34.29 sec	1.25M	1.26M
blocks2_heads2	4.7009	56.64 sec	1.71M	2.53M

blocks2_heads4	4.5017	55.58 sec	1.71M	2.53M
blocks3_heads2	5.7791	33.33 sec	2.18M	3.79M
blocks3_heads4	5.7197	73.39 sec	2.18M	3.79M
blocks4_heads2	5.8426	44.25 sec	2.64M	5.05M
blocks4_heads4	5.8439	48.97 sec	2.64M	5.05M

For the Problem 4 configuration table, the best French-to-English Transformer was blocks1_heads4. It reached 0.1471 word BLEU-4 and 0.3619 character BLEU-4. Exact sequence accuracy was still 0.00%, so the model did not produce perfect full-sentence matches, but BLEU showed stronger partial translation quality. The 1-block models again worked best, while the 3-block and 4-block models performed poorly. This suggests that the small dataset did not support deeper Transformer stacks and that shallow Transformers generalized better.

For the complexity and Homework 3 comparison tables, the best reversed Transformer beat both reversed GRU models from Homework 3 on BLEU-4. The reversed baseline GRU reached 0.0623 word BLEU-4, the reversed attention GRU reached 0.0726, and the best Homework 4 Transformer reached 0.1471. The Transformer also trained faster than the reversed attention GRU in the saved runs. This is an important comparison because it shows that the Transformer helped both translation directions, not only English-to-French. The qualitative table shows the same pattern as Problem 3: the predicted English sentences were not exact matches, but many kept the correct topic or phrase structure. This directly compares the reversed Transformer against the Homework 3 reversed GRU without attention and the reversed GRU with attention.

Model	Direction	Exact	Word BLEU	Char BLEU	Best Val Loss	Total Train Time	Params
Reversed Baseline GRU	French to English	0.00%	0.0623	0.1711	1.3773	202.59 sec	332.7k
Reversed Attention GRU	French to English	0.00%	0.0726	0.1940	1.4748	527.80 sec	666.9k
Homework 4 Best Reversed Transformer	French to English	0.00%	0.1471	0.3619	4.1966	34.29 sec	1.25M

Qualitative Validation

The qualitative table for Problem 4 gives several French validation inputs, the target English sentence, the predicted English sentence, exact-match status, word BLEU-4, and character BLEU-4. These examples show the same pattern as Problem 3. The reversed Transformer usually did not create a perfect sentence, but it often kept the general topic of the target. For example, the generated outputs still referred to related ideas such as pigeons, yoga, walking, jackets, or the bus, even when the exact wording was wrong. This explains why exact accuracy stayed at 0.00%, while BLEU-4 still showed improvement compared with the Homework 3 reversed GRU models.

French Input	Target English	Predicted English	Exact	Word BLEU-4	Char BLEU-4
ils nourrissent les pigeons sur...	they feed the pigeons in the sq...	they feed the ducks in the pond	FAIL	0.2056	0.6483

elle pratique le yoga tous les ...	she practices yoga every morning	she practices the violin every day	FAIL	0.0955	0.5875
j'aime marcher dans la neige	i enjoy walking in the snow	i enjoy walking in the dense fo...	FAIL	0.6148	0.6612
elle adore porter des vestes mo...	she loves to wear modern jackets	she loves to wear colorful hats	FAIL	0.5081	0.5954
le bus de la ville arrive préci...	the city bus arrives precisely ...	the bus arrives at five o'clock	FAIL	0.1905	0.5826

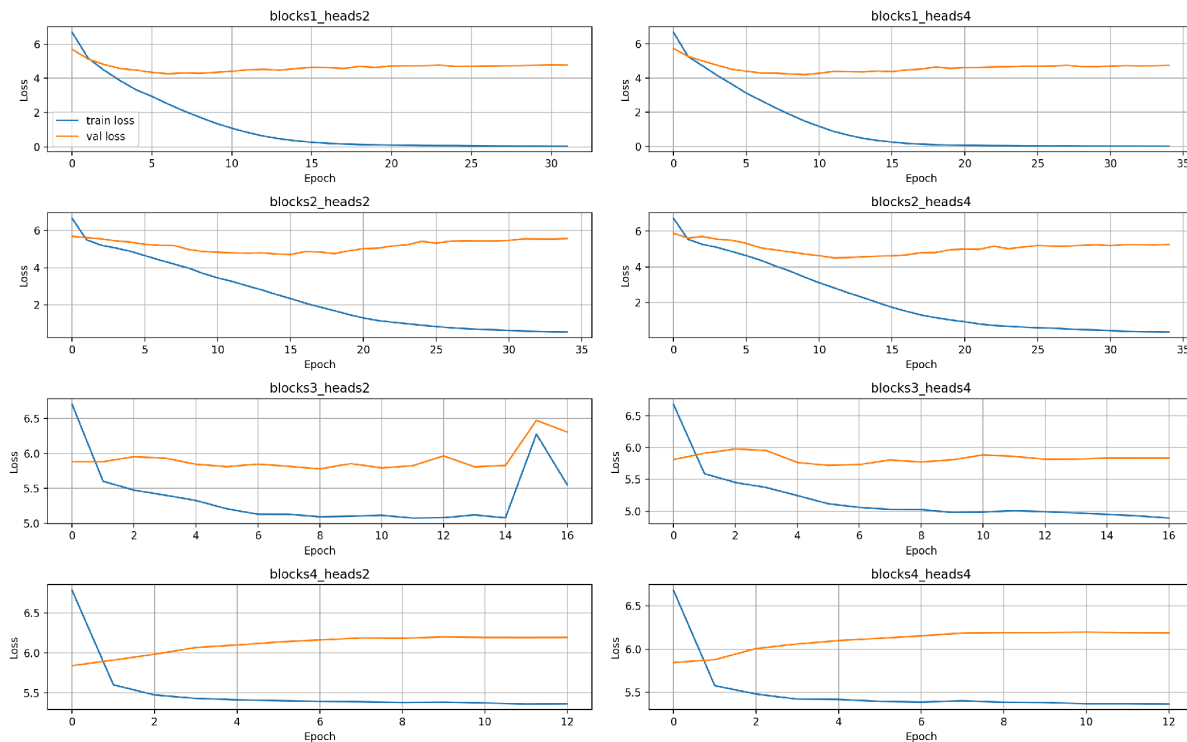
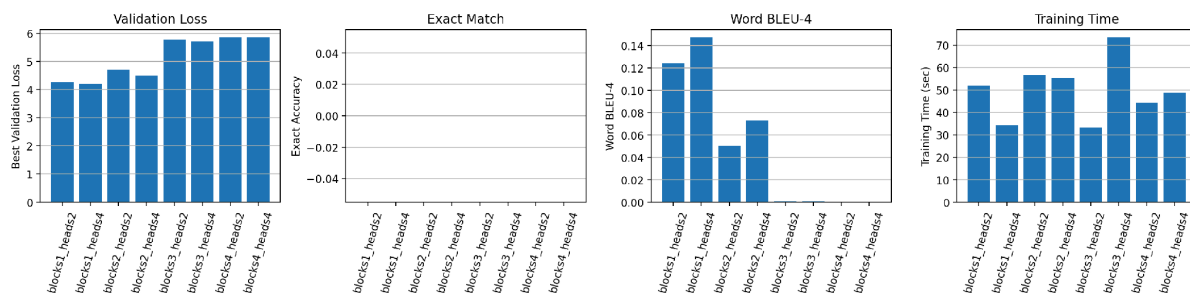


Figure: Problem 4 training and validation loss curves.



The Problem 4 plots show the same depth tradeoff as Problem 3. The shallow 1-block models had the best BLEU scores, while deeper models increased complexity without improving validation translation quality. The loss curves show that training loss dropped, but validation loss stayed higher, so overfitting was still present. The summary plot shows that French-to-English had the highest Transformer word BLEU-4 score, but exact sequence accuracy stayed at

0.00%. Overall, the plots support the conclusion that the reversed task worked best with a smaller Transformer on this small dataset.

Overall Comparison and Conclusion

Best Run	Configuration	Main Accuracy	Word BLEU-4	Total Train Time	Params	FLOPs Proxy
Problem 1 best Transformer	Seq 30	96.60%	N/A	17.23 sec	11.5k	119.0k
Problem 2 best Transformer	base_s30, seq 30	45.51%	N/A	289.96 sec	161.3k	1.45M
Problem 3 best Transformer	blocks1_heads2	0.00%	0.1300	61.90 sec	1.25M	1.26M
Problem 4 best Transformer	blocks1_heads4	0.00%	0.1471	34.29 sec	1.25M	1.26M

Overall, the Transformer performed best on the shorter controlled character-prediction task. In Problem 1, the Homework 4 Transformer with sequence length 30 reached 96.60% best validation accuracy, which was higher than the best Homework 2 recurrent result of 92.10% from the GRU. It also used fewer parameters and a smaller FLOPs proxy. For Problem 2, the trend was different. The Homework 2 LSTM and GRU models still performed better on Tiny Shakespeare, with the LSTM sequence length 20 model giving the best overall balance of validation loss, top-5 accuracy, and perplexity. This shows that the Transformer was strongest on the smaller controlled task, while the recurrent models generalized better on the larger Tiny Shakespeare dataset.

For translation, the Transformer improved over the Homework 3 GRU baseline and attention GRU models in both directions when BLEU-4 was used as the main metric. English-to-French reached 0.1300 word BLEU-4, while French-to-English reached 0.1471 word BLEU-4. Exact sequence accuracy was 0.00% for both directions, so BLEU-4 and the qualitative examples were more useful than exact match. French-to-English was slightly stronger by BLEU-4, but English-to-French had the better validation loss at 3.9253 compared with 4.1966. Overall, the two directions were close, with French-to-English slightly better in generated translation quality and English-to-French slightly smoother in validation loss.

Across Problems 1 - 4, all required Transformer configurations, sequence-length studies, comparisons to prior homework baselines, qualitative translation examples, and computational metrics, including training time, inference time, parameters, and FLOPs proxies, were reported.

Source Files and Artifacts

Homework4_Problem1_Transformer_Long_Sequence.ipynb

Homework4_Problem2_TinyShakespeare_Transformer.ipynb

Homework4_Problem3_Transformer_English_French.ipynb

Homework4_Problem4_Transformer_French_English.ipynb

Results_Problem_1 through Results_Problem_4 folders